```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,
initial-scale=1.0"/>
<title>LibraryOS — Management System</title>
<link
href="https://fonts.googleapis.com/css2?family=Playfair
+Display:wght@600;700&family=DM+Sans:wght@400;500;600&d
isplay=swap" rel="stylesheet"/>
<style>
  *{box-sizing:border-box;margin:0;padding:0}
  body{font-family:'DM Sans',sans-
serif;background:#0a0b15;color:#e8e6ff;min-
height:100vh;display:flex}
  :root{
    --bg:#0a0b15;--sidebar:#0d0e1c;--card:#12131f;--
border:#2a2d4a;--border2:#1e2035;
    --text:#e8e6ff;--muted:#8b8daa;--dim:#5a5d7a;--
accent:#6c63ff;--green:#43d9ad;--orange:#f0a500;--
red:#ff5c5c;--purple:#a78bfa;
  }
  ::-webkit-scrollbar{width:6px}::-webkit-scrollbar-
track{background:var(--card)}::-webkit-scrollbar-
thumb{background:var(--border);border-radius:3px}

  /* SIDEBAR */
  #sidebar{width:220px;background:var(--
sidebar);border-right:1px solid var(--
border2);padding:2rem 1rem;display:flex;flex-
```

```
direction:column;gap:4px;flex-shrink:0;min-
height:100vh}

  .logo{padding:0 0.8rem;margin-bottom:2rem}

  .logo h1{font-family:'Playfair Display',serif;font-
size:1.3rem;color:var(--text);line-height:1.2}

  .logo span{color:#3a3d5a;font-size:0.72rem}

  .nav-btn{display:flex;align-
items:center;gap:12px;padding:0.7rem 0.8rem;border-
radius:10px;border:none;background:none;color:var(--
dim);cursor:pointer;text-align:left;font-
size:0.9rem;font-family:'DM Sans',sans-
serif;width:100%;transition:all 0.15s}

  .nav-btn:hover{color:var(--
purple);background:#6c63ff11}

  .nav-btn.active{color:var(--
purple);background:#6c63ff22}

  .sidebar-stats{margin-
top:auto;padding:0.8rem;background:var(--card);border-
radius:12px;border:1px solid var(--border2)}

  .sidebar-stats p{font-size:0.72rem;text-
transform:uppercase;letter-
spacing:0.05em;color:#3a3d5a;margin-bottom:6px}

  .stat-line{font-size:0.82rem;margin-top:2px}


  /* MAIN */

  #main{flex:1;padding:2.5rem 2rem;overflow-y:auto}

  .page{display:none}.page.active{display:block}


  /* PAGE HEADER */

  .page-header{display:flex;justify-content:space-
between;align-items:flex-start;margin-
bottom:1.5rem;flex-wrap:wrap;gap:12px}
```

```
  .page-title h1{font-family:'Playfair
Display',serif;font-size:1.8rem;color:var(--text)}
  .page-title p{color:var(--dim);margin-top:4px}

  /* BUTTONS */
  .btn{display:inline-flex;align-
items:center;gap:8px;border:none;border-
radius:10px;padding:0.6rem 1.2rem;cursor:pointer;font-
family:'DM Sans',sans-serif;font-size:0.9rem;font-
weight:600;transition:opacity 0.15s}
  .btn:hover{opacity:0.85}
  .btn-primary{background:var(--accent);color:#fff}
  .btn-green{background:var(--green);color:#0a0b15}
  .btn-ghost{background:var(--card);border:1px solid
var(--border);color:var(--muted)}
  .btn-danger{background:#ff5c5c22;color:var(--
red);border:none;border-radius:8px;padding:4px
8px;cursor:pointer;font-family:'DM Sans',sans-serif}
  .btn-edit{background:#6c63ff22;color:var(--
purple);border:none;border-radius:8px;padding:4px
10px;cursor:pointer;font-size:0.8rem;font-family:'DM
Sans',sans-serif}
  .btn-return{display:inline-flex;align-
items:center;gap:6px;background:#43d9ad22;color:var(--
green);border:none;border-radius:8px;padding:4px
10px;cursor:pointer;font-size:0.8rem;font-family:'DM
Sans',sans-serif}

  /* STATS GRID */
  .stats-grid{display:grid;grid-template-
columns:repeat(auto-
fit,minmax(130px,1fr));gap:16px;margin-bottom:2.5rem}
```

```
  .stat-card{background:var(--card);border-radius:14px;padding:1.2rem;text-align:center}
  .stat-card .val{font-size:2rem;font-weight:700;font-family:'Playfair Display',serif}
  .stat-card .lbl{color:var(--dim);font-size:0.8rem;margin-top:4px}

  /* TABLE */
  .table-wrap{background:var(--card);border:1px solid var(--border);border-radius:14px;overflow:hidden;overflow-x:auto}
  table{width:100%;border-collapse:collapse;min-width:600px}
  thead tr{border-bottom:1px solid var(--border)}
  th{padding:0.9rem 1rem;text-align:left;color:var(--dim);font-size:0.78rem;text-transform:uppercase;letter-spacing:0.05em;font-weight:600}
  td{padding:0.85rem 1rem;border-bottom:1px solid var(--border2);font-size:0.88rem}
  tr:last-child td{border-bottom:none}
  .td-title{color:var(--text);font-weight:500}
  .td-muted{color:var(--muted)}
  .td-dim{color:var(--dim);font-size:0.82rem}
  .td-center{text-align:center}

  /* BADGES */
  .badge{border-radius:20px;padding:2px 10px;font-size:0.75rem;font-weight:600}
  .badge-genre{background:#6c63ff22;color:var(--purple)}
```

```
  .badge-issued{background:#f0a50020;color:var(--
orange)}
  .badge-overdue{background:#ff5c5c20;color:var(--red)}
  .badge-returned{background:#43d9ad20;color:var(--
green)}


  /* SEARCH + FILTER */
  .search-row{display:flex;gap:12px;margin-
bottom:1.5rem;flex-wrap:wrap}
  .search-wrap{position:relative;flex:1;min-
width:200px}
  .search-wrap
svg{position:absolute;left:12px;top:50%;transform:trans
lateY(-50%);color:var(--dim)}
  input,select{background:var(--card);border:1px solid
var(--border);border-radius:8px;color:var(--
text);padding:0.6rem 0.8rem;font-
size:0.9rem;outline:none;font-family:'DM Sans',sans-
serif;width:100%}
  .search-wrap input{padding-left:36px}
  select option{background:var(--card);color:var(--
text)}
  input::placeholder{color:var(--dim)}
  input[type=number]{width:100%}
  input:focus,select:focus{border-color:var(--accent)}


  /* FILTER TABS */
  .filter-tabs{display:flex;gap:8px;margin-
bottom:1.5rem;flex-wrap:wrap}
  .filter-tab{background:var(--card);color:var(--
dim);border:1px solid var(--border);border-
```

```
radius:20px;padding:5px 16px;cursor:pointer;font-
size:0.82rem;font-family:'DM Sans',sans-serif;text-
transform:capitalize;transition:all 0.15s}

  .filter-tab.active{background:var(--
accent);color:#fff;border-color:var(--accent)}


  /* MEMBER CARDS */

  .members-grid{display:grid;grid-template-
columns:repeat(auto-fill,minmax(270px,1fr));gap:16px}

  .member-card{background:var(--card);border:1px solid
var(--border);border-radius:14px;padding:1.3rem}

  .member-card-top{display:flex;justify-content:space-
between;align-items:flex-start;margin-bottom:0.8rem}

  .avatar{width:44px;height:44px;border-
radius:50%;background:linear-
gradient(135deg,#6c63ff,#a78bfa);display:flex;align-
items:center;justify-content:center;color:#fff;font-
weight:700;font-size:1.1rem;flex-shrink:0}

  .member-name{color:var(--text);font-
weight:600;margin-bottom:4px}

  .member-info{color:var(--dim);font-size:0.82rem}

  .member-footer{margin-top:10px;padding-
top:10px;border-top:1px solid var(--
border2);color:var(--dim);font-size:0.75rem}


  /* MODAL */

  #modal-
overlay{display:none;position:fixed;inset:0;background:
rgba(10,10,20,0.75);z-index:100;align-
items:center;justify-content:center;backdrop-
filter:blur(4px)}

  #modal-overlay.open{display:flex}
```

```
  .modal{background:var(--card);border:1px solid var(--
border);border-radius:16px;width:min(520px,95vw);max-
height:90vh;overflow-y:auto;padding:2rem;box-shadow:0
24px 60px rgba(0,0,0,0.6)}

  .modal-header{display:flex;justify-content:space-
between;align-items:center;margin-bottom:1.5rem}

  .modal-title{font-family:'Playfair
Display',serif;color:var(--text);font-size:1.3rem}

  .modal-close{background:none;border:none;color:var(--
dim);cursor:pointer;font-size:1.3rem;line-height:1}

  .field{margin-bottom:1rem}

  .field label{display:block;color:var(--muted);font-
size:0.78rem;margin-bottom:6px;letter-
spacing:0.05em;text-transform:uppercase}

  .modal-footer{display:flex;gap:12px;justify-
content:flex-end;margin-top:1.5rem}

  .info-box{background:#1c1e30;border-
radius:10px;padding:0.8rem 1rem;margin-
bottom:1rem;color:var(--muted);font-size:0.83rem}


  /* EMPTY STATE */

  .empty{color:var(--dim);text-
align:center;padding:2rem}


  /* RECENT LIST */

  .recent-list{background:var(--card);border:1px solid
var(--border);border-radius:14px;overflow:hidden}

  .recent-item{display:flex;align-
items:center;padding:0.9rem 1.2rem;border-bottom:1px
solid var(--border2);gap:16px}

  .recent-item:last-child{border-bottom:none}
```

```
  .recent-item .ri-title{color:var(--text);font-size:0.9rem}
  .recent-item .ri-sub{color:var(--dim);font-size:0.78rem}

  svg{display:inline-block;vertical-align:middle}
</style>
</head>
<body>

<!-- SIDEBAR -->
<aside id="sidebar">
  <div class="logo">
    <h1>📚 LibraryOS</h1>
    <span>Management System</span>
  </div>
  <button class="nav-btn active" onclick="navigate('dashboard')">
    <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><rect x="3" y="3" width="7" height="7"/><rect x="14" y="3" width="7" height="7"/><rect x="14" y="14" width="7" height="7"/><rect x="3" y="14" width="7" height="7"/></svg>
    Dashboard
  </button>
  <button class="nav-btn" onclick="navigate('books')">
    <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5
```

```
17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4
19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
    Books
  </button>
  <button class="nav-btn"
onclick="navigate('members')">
    <svg width="16" height="16" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-
width="2"><path d="M17 21v-2a4 4 0 0 0-4-4H5a4 4 0 0 0-
4 4v2"/><circle cx="9" cy="7" r="4"/><path d="M23 21v-
2a4 4 0 0 0-3-3.87"/><path d="M16 3.13a4 4 0 0 1 0
7.75"/></svg>
    Members
  </button>
  <button class="nav-btn"
onclick="navigate('transactions')">
    <svg width="16" height="16" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-
width="2"><line x1="5" y1="12" x2="19"
y2="12"/><polyline points="12 5 19 12 12 19"/></svg>
    Transactions
  </button>
  <div class="sidebar-stats">
    <p>Quick Stats</p>
    <div class="stat-line" style="color:var(--green)"
id="stat-available">✓ — books available</div>
    <div class="stat-line" style="color:var(--red)"
id="stat-overdue">⚠ — overdue</div>
  </div>
</aside>
```

```
<!-- MAIN -->
<main id="main">

  <!-- DASHBOARD -->
  <div id="page-dashboard" class="page active">
    <div class="page-header"><div class="page-title"><h1>Dashboard</h1><p>Library overview at a glance</p></div></div>
    <div class="stats-grid" id="dash-stats"></div>
    <h3 style="font-family:'Playfair Display',serif;margin-bottom:1rem;color:var(--text)">Recent Transactions</h3>
    <div class="recent-list" id="dash-recent"></div>
  </div>

  <!-- BOOKS -->
  <div id="page-books" class="page">
    <div class="page-header">
      <div class="page-title"><h1>Books</h1><p>Manage your library catalog</p></div>
      <button class="btn btn-primary" onclick="openBookModal()">
        <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><line x1="12" y1="5" x2="12" y2="19"/><line x1="5" y1="12" x2="19" y2="12"/></svg>
        Add Book
      </button>
    </div>
    <div class="search-row">
```

```
      <div class="search-wrap">
        <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="11" cy="11" r="8"/><line x1="21" y1="21" x2="16.65" y2="16.65"/></svg>
        <input type="text" id="book-search" placeholder="Search title, author, ISBN..." oninput="renderBooks()"/>
      </div>
      <select id="genre-filter" onchange="renderBooks()" style="width:auto">
        <option value="All">All Genres</option>
<option>Fiction</option><option>Technology</option><option>Sci-Fi</option>
<option>History</option><option>Science</option><option>Biography</option><option>Other</option>
      </select>
    </div>
    <div class="table-wrap">
      <table><thead><tr>
<th>Title</th><th>Author</th><th>Genre</th><th>ISBN</th><th class="td-center">Copies</th><th class="td-center">Available</th><th>Actions</th>
      </tr></thead>
      <tbody id="books-tbody"></tbody></table>
    </div>
  </div>
```

```
  <!-- MEMBERS -->
  <div id="page-members" class="page">
    <div class="page-header">
      <div class="page-
title"><h1>Members</h1><p>Library membership
directory</p></div>
      <button class="btn btn-primary"
onclick="openMemberModal()">
        <svg width="16" height="16" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-
width="2"><line x1="12" y1="5" x2="12" y2="19"/><line
x1="5" y1="12" x2="19" y2="12"/></svg>
        Add Member
      </button>
    </div>
    <div class="search-row">
      <div class="search-wrap">
        <svg width="16" height="16" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-
width="2"><circle cx="11" cy="11" r="8"/><line x1="21"
y1="21" x2="16.65" y2="16.65"/></svg>
        <input type="text" id="member-search"
placeholder="Search name, email, phone..."
oninput="renderMembers()"/>
      </div>
    </div>
    <div class="members-grid" id="members-grid"></div>
  </div>


  <!-- TRANSACTIONS -->
```

```
  <div id="page-transactions" class="page">
    <div class="page-header">
      <div class="page-
title"><h1>Transactions</h1><p>Issue and return
books</p></div>
      <button class="btn btn-green"
onclick="openIssueModal()">
        <svg width="16" height="16" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-
width="2"><line x1="12" y1="5" x2="12" y2="19"/><line
x1="5" y1="12" x2="19" y2="12"/></svg>
        Issue Book
      </button>
    </div>
    <div class="filter-tabs" id="tx-filters">
      <button class="filter-tab active"
onclick="setTxFilter('all',this)">All</button>
      <button class="filter-tab"
onclick="setTxFilter('issued',this)">Issued</button>
      <button class="filter-tab"
onclick="setTxFilter('overdue',this)">Overdue</button>
      <button class="filter-tab"
onclick="setTxFilter('returned',this)">Returned</button
>
    </div>
    <div class="table-wrap">
      <table><thead><tr>
        <th>#</th><th>Book</th><th>Member</th><th>Issue
Date</th><th>Due Date</th><th>Return
Date</th><th>Status</th><th>Action</th>
      </tr></thead>
```

```
      <tbody id="tx-tbody"></tbody></table>
    </div>
  </div>

</main>

<!-- MODAL -->
<div id="modal-overlay" onclick="closeModal(event)">
  <div class="modal">
    <div class="modal-header">
      <span class="modal-title" id="modal-
title">Modal</span>
      <button class="modal-close"
onclick="closeModalDirect()">X</button>
    </div>
    <div id="modal-body"></div>
  </div>
</div>

<script>
// ==================== DATA ====================
let books = [
  {id:1,title:"The Great Gatsby",author:"F. Scott
Fitzgerald",genre:"Fiction",isbn:"978-0-7432-7356-
5",total:3,available:2},
  {id:2,title:"Clean Code",author:"Robert C.
Martin",genre:"Technology",isbn:"978-0-13-235088-
4",total:5,available:4},
```

```
  {id:3,title:"Dune",author:"Frank Herbert",genre:"Sci-Fi",isbn:"978-0-441-17271-9",total:2,available:1},
  {id:4,title:"To Kill a Mockingbird",author:"Harper Lee",genre:"Fiction",isbn:"978-0-06-112008-4",total:4,available:4},
  {id:5,title:"Introduction to Algorithms",author:"Cormen et al.",genre:"Technology",isbn:"978-0-26-204630-5",total:3,available:0},
];
let members = [
  {id:1,name:"Harshali Patil",email:"harshali@example.com",phone:"9876543210",joinDate:"2024-01-15"},
  {id:2,name:"Riya Sharma",email:"riya@example.com",phone:"9123456789",joinDate:"2024-03-10"},
  {id:3,name:"Arjun Mehta",email:"arjun@example.com",phone:"9988776655",joinDate:"2024-06-01"},
];
let transactions = [
  {id:1,bookId:3,memberId:2,issueDate:"2025-04-01",dueDate:"2025-04-15",returnDate:null,status:"issued"},
  {id:2,bookId:5,memberId:1,issueDate:"2025-03-20",dueDate:"2025-04-03",returnDate:null,status:"overdue"},
  {id:3,bookId:1,memberId:3,issueDate:"2025-03-10",dueDate:"2025-03-24",returnDate:"2025-03-22",status:"returned"},
];
let txFilter = "all";
```

```
let editingBookId = null;
let editingMemberId = null;

// ==================== HELPERS ====================
function today(){ return new
Date().toISOString().split("T")[0]; }
function dueDate(){ const d=new Date();
d.setDate(d.getDate()+14); return
d.toISOString().split("T")[0]; }
function isOverdue(d){ return d && new Date(d)<new
Date(); }
function nextId(arr){ return
arr.length===0?1:Math.max(...arr.map(x=>x.id))+1; }
function statusBadge(s){
  const map={issued:{cls:"badge-
issued",txt:"Issued"},overdue:{cls:"badge-
overdue",txt:"Overdue"},returned:{cls:"badge-
returned",txt:"Returned"}};
  const m=map[s]||{cls:"",txt:s};
  return `<span class="badge
${m.cls}">${m.txt}</span>`;
}

// ==================== NAVIGATION ====================
function navigate(page){

document.querySelectorAll('.page').forEach(p=>p.classLi
st.remove('active'));
  document.querySelectorAll('.nav-
btn').forEach(b=>b.classList.remove('active'));
```

```
  document.getElementById('page-
'+page).classList.add('active');

  document.querySelectorAll('.nav-
btn')[['dashboard','books','members','transactions'].in
dexOf(page)].classList.add('active');

  if(page==='dashboard') renderDashboard();

  if(page==='books') renderBooks();

  if(page==='members') renderMembers();

  if(page==='transactions') renderTransactions();

}


// ==================== AUTO OVERDUE
====================

function checkOverdue(){

  transactions=transactions.map(t=>{

    if((t.status==='issued')&&isOverdue(t.dueDate))
return {...t,status:'overdue'};

    return t;

  });

}


// ==================== SIDEBAR STATS
====================

function updateSidebarStats(){

  document.getElementById('stat-
available').textContent='✓
'+books.reduce((s,b)=>s+b.available,0)+' books
available';

  document.getElementById('stat-
overdue').textContent='⚠
```

```
'+transactions.filter(t=>t.status==='overdue').length+'
overdue';
}

// ==================== DASHBOARD ====================
function renderDashboard(){
  const total=books.reduce((s,b)=>s+b.total,0);
  const avail=books.reduce((s,b)=>s+b.available,0);
  const
issued=transactions.filter(t=>t.status==='issued'||t.st
atus==='overdue').length;
  const
overdue=transactions.filter(t=>t.status==='overdue').le
ngth;
  const stats=[
    {label:'Total Books',value:total,color:'#6c63ff'},
    {label:'Available',value:avail,color:'#43d9ad'},
    {label:'Issued',value:issued,color:'#f0a500'},
    {label:'Overdue',value:overdue,color:'#ff5c5c'},

{label:'Members',value:members.length,color:'#a78bfa'},
  ];
  document.getElementById('dash-
stats').innerHTML=stats.map(s=>`
    <div class="stat-card" style="border:1px solid
${s.color}33">
      <div class="val"
style="color:${s.color}">${s.value}</div>
      <div class="lbl">${s.label}</div>
    </div>`).join('');
```

```
  const recent=[...transactions].reverse().slice(0,6);
  const el=document.getElementById('dash-recent');
  if(recent.length===0){el.innerHTML='<p
class="empty">No transactions yet.</p>';return;}
  el.innerHTML=recent.map(t=>{
    const b=books.find(x=>x.id===t.bookId);
    const m=members.find(x=>x.id===t.memberId);
    return `<div class="recent-item">
      <div style="flex:1">
        <div class="ri-
title">${b?b.title:'Unknown'}</div>
        <div class="ri-sub">${m?m.name:'Unknown'} ·
${t.issueDate}</div>
      </div>
      ${statusBadge(t.status)}
    </div>`;
  }).join('');
  updateSidebarStats();
}

// ===================== BOOKS =====================
function renderBooks(){
  const q=(document.getElementById('book-
search').value||'').toLowerCase();
  const g=document.getElementById('genre-
filter').value;
  const filtered=books.filter(b=>{
```

```
    const mq=b.title.toLowerCase().includes(q)||b.author.toLowerCase().includes(q)||b.isbn.includes(q);
    const mg=g==='All'||b.genre===g;
    return mq&&mg;
  });
  const tbody=document.getElementById('books-tbody');
  if(filtered.length===0){tbody.innerHTML='<tr><td colspan="7" class="empty">No books found.</td></tr>';return;}
  tbody.innerHTML=filtered.map(b=>`
    <tr>
      <td class="td-title">${b.title}</td>
      <td class="td-muted">${b.author}</td>
      <td><span class="badge badge-genre">${b.genre}</span></td>
      <td class="td-dim">${b.isbn}</td>
      <td class="td-center td-muted">${b.total}</td>
      <td class="td-center" style="color:${b.available===0?'var(--red)':b.available<=1?'var(--orange)':'var(--green)'};font-weight:700">${b.available}</td>
      <td>
        <button class="btn-edit" onclick="openBookModal(${b.id})">Edit</button>
        <button class="btn-danger" onclick="deleteBook(${b.id})">🗑</button>
      </td>
    </tr>`).join('');
  updateSidebarStats();
```

```
}

function openBookModal(id=null){
  editingBookId=id;
  const b=id?books.find(x=>x.id===id):null;
  document.getElementById('modal-
title').textContent=id?'Edit Book':'Add New Book';
  document.getElementById('modal-body').innerHTML=`
    <div class="field"><label>Title</label><input
id="f-title" value="${b?b.title:''}" placeholder="Book
title"/></div>
    <div class="field"><label>Author</label><input
id="f-author" value="${b?b.author:''}"
placeholder="Author name"/></div>
    <div class="field"><label>Genre</label>
      <select id="f-genre">
        ${['Fiction','Technology','Sci-
Fi','History','Science','Biography','Other'].map(g=>`<o
ption
${b&&b.genre===g?'selected':''}>${g}</option>`).join('
')}
      </select>
    </div>
    <div class="field"><label>ISBN</label><input id="f-
isbn" value="${b?b.isbn:''}" placeholder="978-x-xxx-
xxxxx-x"/></div>
    <div class="field"><label>Total
Copies</label><input type="number" id="f-total" min="1"
value="${b?b.total:1}"/></div>
    <div class="modal-footer">
```

```
      <button class="btn btn-ghost"
onclick="closeModalDirect()">Cancel</button>
      <button class="btn btn-primary"
onclick="saveBook()">Save</button>
    </div>`;
  openModal();
}


function saveBook(){
  const title=document.getElementById('f-
title').value.trim();
  const author=document.getElementById('f-
author').value.trim();
  if(!title||!author){alert('Title and Author are
required!');return;}
  const genre=document.getElementById('f-genre').value;
  const isbn=document.getElementById('f-
isbn').value.trim();
  const total=parseInt(document.getElementById('f-
total').value)||1;
  if(editingBookId){
    books=books.map(b=>{
      if(b.id!==editingBookId) return b;
      const diff=total-b.total;
      return
{...b,title,author,genre,isbn,total,available:Math.max(
0,b.available+diff)};
    });
  } else {
```

```
books.push({id:nextId(books),title,author,genre,isbn,total,available:total});
  }
  closeModalDirect();renderBooks();
}

function deleteBook(id){
  if(!confirm('Delete this book?')) return;
  books=books.filter(b=>b.id!==id);
  renderBooks();
}

// ==================== MEMBERS ====================
function renderMembers(){
  const q=(document.getElementById('member-search').value||'').toLowerCase();
  const filtered=members.filter(m=>m.name.toLowerCase().includes(q)||m.email.toLowerCase().includes(q)||m.phone.includes(q));
  const grid=document.getElementById('members-grid');
  if(filtered.length===0){grid.innerHTML='<p class="empty" style="grid-column:1/-1">No members found.</p>';return;}
  grid.innerHTML=filtered.map(m=>`
    <div class="member-card">
      <div class="member-card-top">
        <div class="avatar">${m.name.charAt(0)}</div>
        <div style="display:flex;gap:8px">
```

```
          <button class="btn-edit"
onclick="openMemberModal(${m.id})">Edit</button>
          <button class="btn-danger"
onclick="deleteMember(${m.id})">🗑</button>
        </div>
      </div>
      <div class="member-name">${m.name}</div>
      <div class="member-info">${m.email}</div>
      <div class="member-info">${m.phone}</div>
      <div class="member-footer">Joined: ${m.joinDate}
· ID: #${m.id}</div>
    </div>`).join('');
}

function openMemberModal(id=null){
  editingMemberId=id;
  const m=id?members.find(x=>x.id===id):null;
  document.getElementById('modal-
title').textContent=id?'Edit Member':'Add New Member';
  document.getElementById('modal-body').innerHTML=`
    <div class="field"><label>Full Name</label><input
id="m-name" value="${m?m.name:''}"
placeholder="Member's full name"/></div>
    <div class="field"><label>Email</label><input
type="email" id="m-email" value="${m?m.email:''}"
placeholder="email@example.com"/></div>
    <div class="field"><label>Phone</label><input
id="m-phone" value="${m?m.phone:''}" placeholder="10-
digit number"/></div>
```

```
    <div class="field"><label>Join Date</label><input
type="date" id="m-join"
value="${m?m.joinDate:today()}"/></div>

    <div class="modal-footer">

      <button class="btn btn-ghost"
onclick="closeModalDirect()">Cancel</button>

      <button class="btn btn-primary"
onclick="saveMember()">Save</button>

    </div>`;

  openModal();

}


function saveMember(){

  const name=document.getElementById('m-
name').value.trim();

  const email=document.getElementById('m-
email').value.trim();

  if(!name||!email){alert('Name and Email are
required!');return;}

  const phone=document.getElementById('m-
phone').value.trim();

  const joinDate=document.getElementById('m-
join').value||today();

  if(editingMemberId){


members=members.map(m=>m.id===editingMemberId?{...m,nam
e,email,phone,joinDate}:m);

  } else {


members.push({id:nextId(members),name,email,phone,joinD
ate});
```

```
  }
  closeModalDirect();renderMembers();
}

function deleteMember(id){
  if(!confirm('Delete this member?')) return;
  members=members.filter(m=>m.id!==id);
  renderMembers();
}

// ==================== TRANSACTIONS
====================
function setTxFilter(f,btn){
  txFilter=f;
  document.querySelectorAll('.filter-
tab').forEach(b=>b.classList.remove('active'));
  btn.classList.add('active');
  renderTransactions();
}

function renderTransactions(){
  const
filtered=(txFilter==='all'?transactions:transactions.fi
lter(t=>t.status===txFilter));
  const tbody=document.getElementById('tx-tbody');
  if(filtered.length===0){tbody.innerHTML='<tr><td
colspan="8" class="empty">No transactions
found.</td></tr>';return;}
  tbody.innerHTML=[...filtered].reverse().map(t=>{
```

```
    const b=books.find(x=>x.id===t.bookId);
    const m=members.find(x=>x.id===t.memberId);
    const
actionBtn=(t.status===='issued'||t.status===='overdue')
      ?`<button class="btn-return"
onclick="returnBook(${t.id})">✓ Return</button>`:'';
    return `<tr>
      <td class="td-dim">#${t.id}</td>
      <td class="td-title" style="max-
width:150px;overflow:hidden;text-
overflow:ellipsis;white-
space:nowrap">${b?b.title:'?'}</td>
      <td class="td-muted">${m?m.name:'?'}</td>
      <td class="td-dim">${t.issueDate}</td>
      <td style="color:${t.status===='overdue'?'var(--
red)':'var(--dim)'};font-
size:0.82rem">${t.dueDate}</td>
      <td class="td-dim">${t.returnDate||'—'}</td>
      <td>${statusBadge(t.status)}</td>
      <td>${actionBtn}</td>
    </tr>`;
  }).join('');
  updateSidebarStats();
}

function openIssueModal(){
  const available=books.filter(b=>b.available>0);
  document.getElementById('modal-
title').textContent='Issue a Book';
```

```
  document.getElementById('modal-body').innerHTML=`
    <div class="field"><label>Select Book</label>
      <select id="i-book">
        <option value="">-- Choose a book --</option>
        ${available.map(b=>`<option
value="${b.id}">${b.title} (${b.available}
left)</option>`).join('')}
      </select>
    </div>
    <div class="field"><label>Select Member</label>
      <select id="i-member">
        <option value="">-- Choose a member --</option>
        ${members.map(m=>`<option
value="${m.id}">${m.name}</option>`).join('')}
      </select>
    </div>
    <div class="info-box">🕐 Issue date: ${today()} ·
Due date: ${dueDate()} (14 days)</div>
    <div class="modal-footer">
      <button class="btn btn-ghost"
onclick="closeModalDirect()">Cancel</button>
      <button class="btn btn-green"
onclick="issueBook()">Issue Book</button>
    </div>`;
  openModal();
}

function issueBook(){
```

```
  const bookId=parseInt(document.getElementById('i-
book').value);

  const memberId=parseInt(document.getElementById('i-
member').value);

  if(!bookId||!memberId){alert('Please select both a
book and a member!');return;}

  const book=books.find(b=>b.id===bookId);

  if(!book||book.available===0){alert('Book not
available!');return;}


transactions.push({id:nextId(transactions),bookId,membe
rId,issueDate:today(),dueDate:dueDate(),returnDate:null
,status:'issued'});


books=books.map(b=>b.id===bookId?{...b,available:b.avai
lable-1}:b);

  closeModalDirect();renderTransactions();

}


function returnBook(txId){

  const tx=transactions.find(t=>t.id===txId);

  if(!tx) return;


transactions=transactions.map(t=>t.id===txId?{...t,retu
rnDate:today(),status:'returned'}:t);


books=books.map(b=>b.id===tx.bookId?{...b,available:b.a
vailable+1}:b);

  renderTransactions();

}
```

```
// ==================== MODAL ====================

function openModal(){ document.getElementById('modal-
overlay').classList.add('open'); }

function closeModalDirect(){
document.getElementById('modal-
overlay').classList.remove('open'); }

function closeModal(e){
if(e.target===document.getElementById('modal-overlay'))
closeModalDirect(); }


// ==================== INIT ====================

checkOverdue();

renderDashboard();

updateSidebarStats();

</script>

</body>

</html>
```